

SAP CPI Project Documentation

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Scenario: Product Shipment Orders Processing

1.Scenario Overview

This integration handles **bulk shipment orders** by splitting them into individual orders, validating quantities, routing them based on region, and converting them into structured formats for warehouse systems. The scenario demonstrates the use of **Splitter**, **Filter**, **Router**, and **Converter** palette steps in SAP CPI and helps streamline logistics automation.

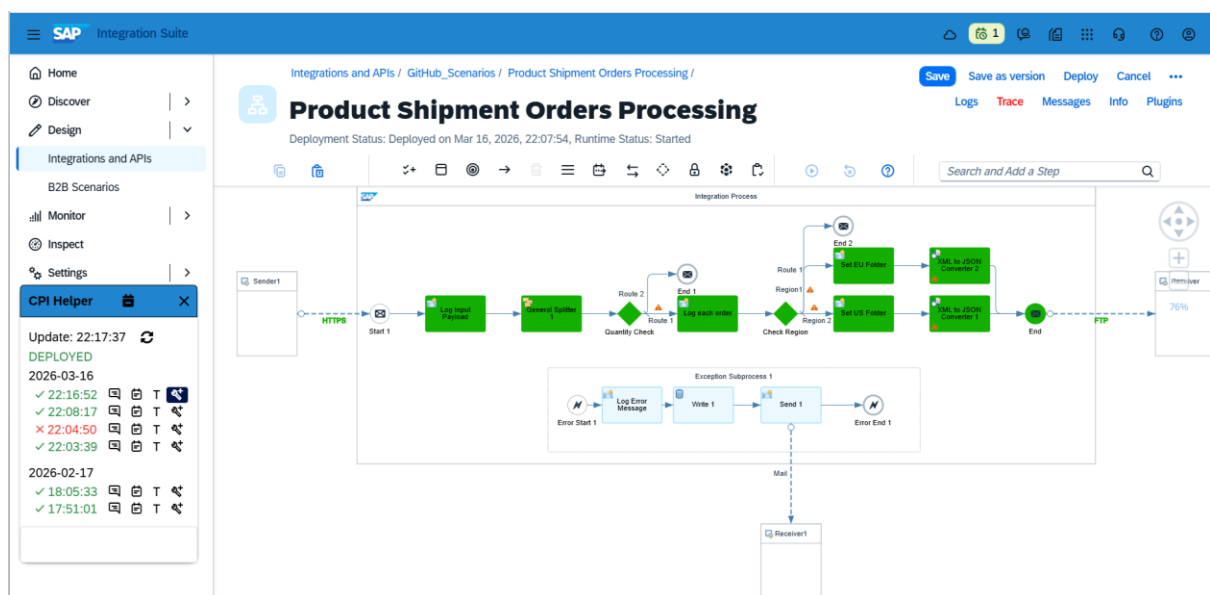
Business Use Case

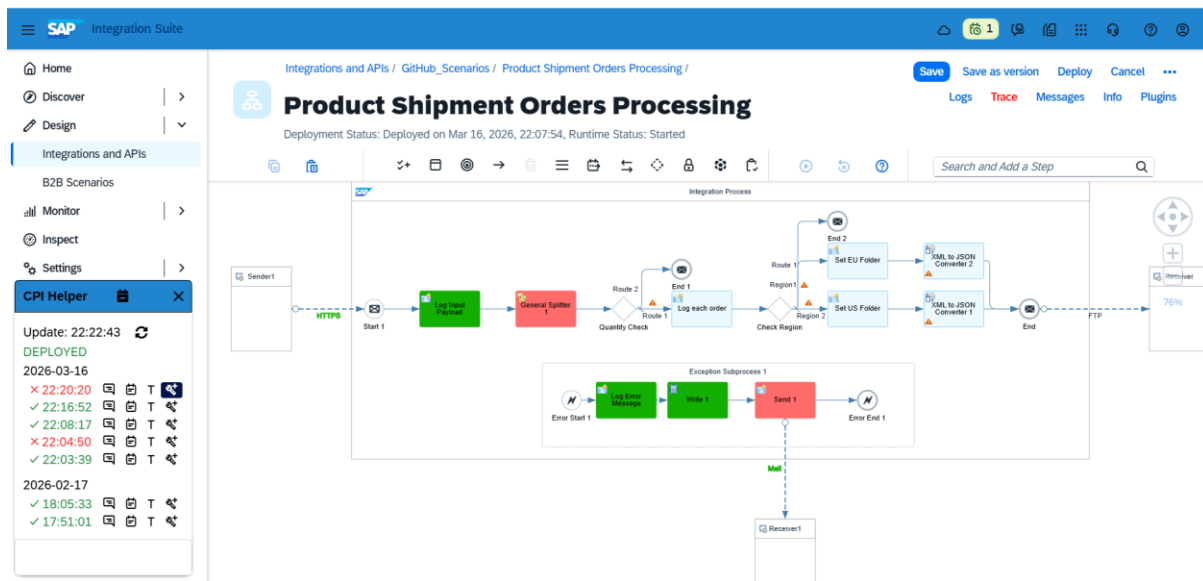
A logistics system sends a **daily shipment file** containing multiple product orders for different regional warehouses (EU, US). Each order includes the product ID, quantity, and region.

Processing needs to:

- Validate orders (quantity > 0)
- Route based on region
- Convert format from XML to JSON
- Store orders as separate files in **region-specific folders**

2.Architechture Diagram





3. Integration Flow

Steps

1. HTTP picks up batch XML file

The screenshot shows the configuration for the HTTP receiver in the SAP Integration Suite. The 'Connection' tab is selected, and the 'REQUEST PROCESSING' section is visible. The configuration is as follows:

- Address:** /ProductShipment/Orders
- Authorization:** User Role
- User Role:** ESBMessaging.send
- CSRF Protected:** ☒

2. XML Splitter splits each <Order> from the batch
3. Filter removes invalid orders (e.g., quantity = 0)
4. Content Modifier enriches each message with metadata (e.g., processed timestamp)

5. **Router** directs orders to regional branches (EU/US)

Router

General **Processing**

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	Route 2		☒
2	Route 1	//Quantity > '0'	☐

Router

General **Processing**

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	Region 2	//Region = 'US'	☐
2	Region1	//Region = 'EU'	☐
3	Route 1		☒

6. **XML to JSON Converter** transforms each valid order

7. **File Receiver Adapter** writes to /shipments/EU/ or /shipments/US/

8. **Sample Input XML (Batch Shipment File)**

<Shipments>

<Order>

<ID>ORD001</ID>

<Region>EU</Region>

<Quantity>10</Quantity>

</Order>

<Order>

<ID>ORD002</ID>

<Region>US</Region>

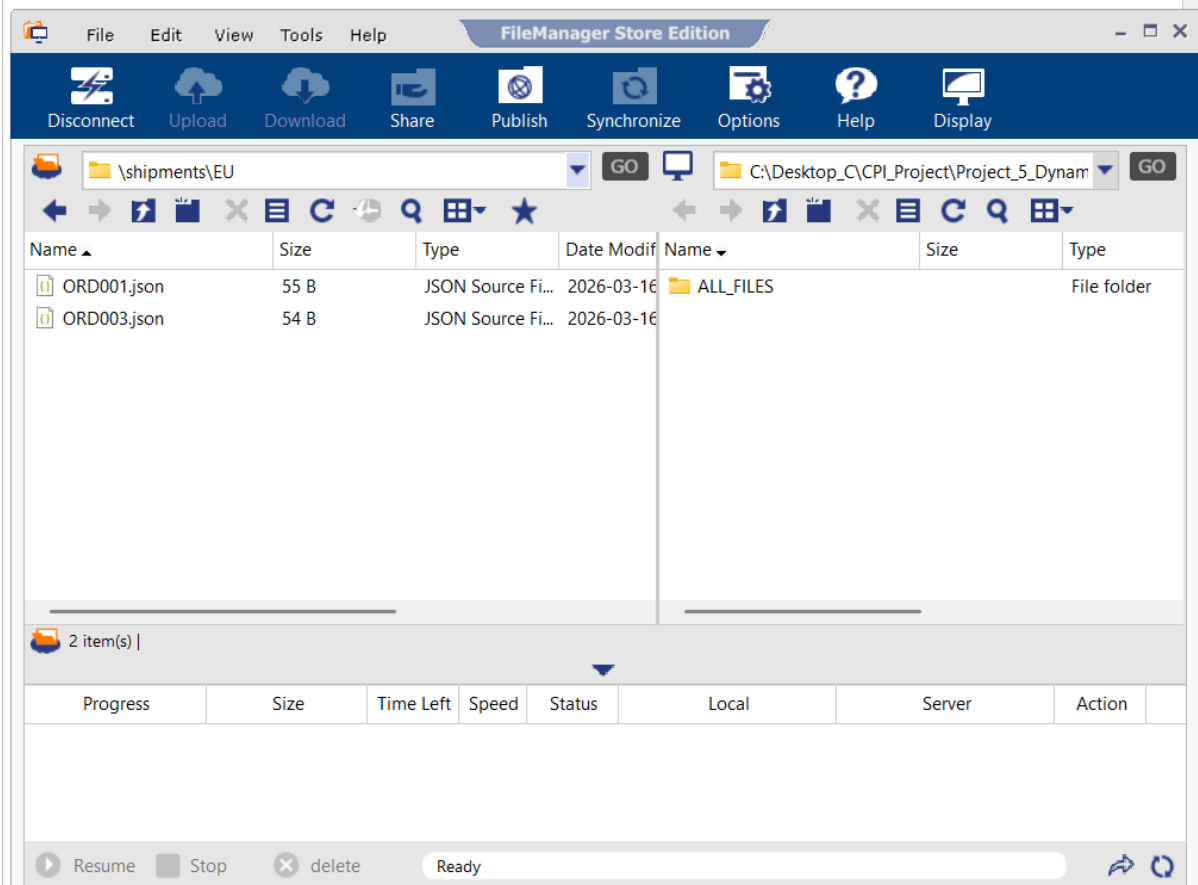
```
        <Quantity>0</Quantity>
    </Order>
    <Order>
        <ID>ORD003</ID>
        <Region>EU</Region>
        <Quantity>5</Quantity>
    </Order>
</Shipments>
```

Processing Example

- **Filtered out:** ORD002 (Quantity = 0)
- **Processed and routed:** ORD001 → EU folder, ORD003 → EU folder

9. JSON Output Example (ORD001)

```
{
  "Order": {
    "ID": "ORD001",
    "Region": "EU",
    "Quantity": 10
  }
}
```



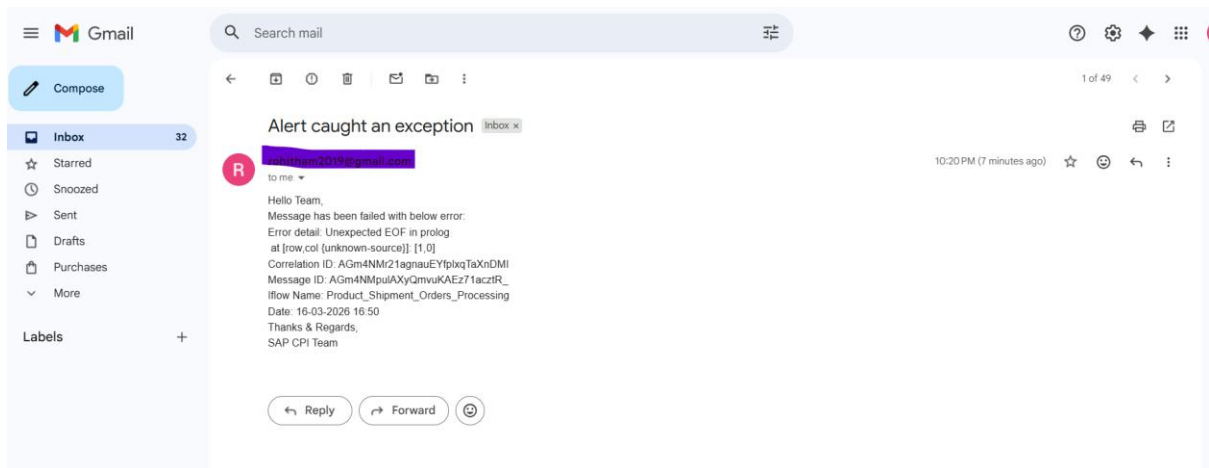
Files stored in HR system

File Output (File Receiver Adapter)

- Output path: `/shipments/EU/ORD001.json` `/shipments/EU/ORD003.json`
- Output format: JSON
- File naming: Based on order ID and region

10. Exception Handling

- **Catch step errors** in file write or transformation
- Log to **Data Store**
- Notify logistics support team via **Gmail Adapter**



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Iflow Package



Product Shipment Orders Processing.zip